

WILLIAM KIRSCHENMAN

Ph.D. Student, Department of Industrial and Systems Engineering, North Carolina State University

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EDUCATION

Ph.D. (<i>In progress</i>), Operations Research, <i>North Carolina State University</i>	Anticipated 2026
M.S. , Operations Research, <i>George Mason University</i>	2020
MPA , Public Administration, <i>Webster University</i>	2015
B.S. , Mechanical Engineering, <i>United States Military Academy</i>	2010

PROFESSIONAL MEMBERSHIPS

Military Operations Research Society (MORS)	2021 - Present
The Institute for Operations Research and the Management Sciences (INFORMS)	2023 - Present
Engineer Intern in New York	2010

FELLOWSHIPS AND AWARDS

Salah E. Elmaghraby Distinguished Student Award, Operations Research Program, NC State University	2025
General Omar N. Bradley Research Fellowship in Mathematics	2024, 2025
Army Operations Research Symposium (AORS) Best Presentation, Wargaming Working Group	2021
Outstanding Academic Achievement Award in Operations Research, George Mason University	2020

WORK EXPERIENCE

The Research and Analysis Center (TRAC), Army Futures Command (AFC)	August 2020 - May 2023
<i>Combat Operations Analyst</i>	<i>White Sands Missile Range, NM</i>

- Lead developer of TRAC's Visualization Application with Data collection in R (VADR), a cloud-based planning and analytical tool for wargaming and other data collection events.
- Led planning, execution and analysis for AFC's 2020 Capstone Wargame focused on the U.S. European Command area of responsibility.
- Planned and executed warfighter touchpoints, operational unit focus groups, and a strategic commanders assessment for AFC's Project Convergence 22, a Combined Joint Force experiment informing Joint Warfighting Concept, Joint All Domain Command and Control.

2nd Infantry Brigade Combat Team, 4th Infantry Division	August 2015 - December 2018
<i>Company Commander, Brigade Engineer Plans Officer</i>	<i>Fort Carson, CO</i>

- Commanded a company of 101 Light Engineer (Sapper) Soldiers responsible for organizational and installation equipment with combined value in excess of \$20 million (12 months).
- Commanded a battalion headquarters company of over 100 Soldiers and responsible for organizational and installation equipment with combined value in excess of \$15 million (12 months).
- Served as the primary engineer planner for an Infantry Brigade Combat Team of seven battalions and over 3,500 Soldiers.
- Served as the Train, Advise, Assist Command – South Combined Joint Engineer Section's Operations Officer, responsible for planning and gaining funding approval for all non-MILCON construction projects in southern Afghanistan.

G-3 Engineer Section, 2nd Infantry Division	July 2014 - July 2015
<i>Operations Officer</i>	<i>Camp Red Cloud, Republic of Korea (ROK)</i>

- Planned, synchronized and assessed engineer operations for the ROK – U.S. Combined 2nd Infantry Division.

Security Force Assist/Advise Team (SFAAT) #2

December 2012 - September 2013

*Engineer Company Commander Advisor**Logar Province, Afghanistan*

- Advised an Afghan National Army (ANA) Engineer Company Commander in Logar Province, Afghanistan, over the course of nine months in all areas pertaining to command.
- Implemented demolitions and horizontal engineering courses for ANA engineers, resulting in the highest improvised explosive device (IED) find rate in the ANA Corps, along with self-sustaining combat outpost construction operations.

11th Engineer Battalion, 20th Engineer Brigade

February 2011 - December 2013

*Company Executive Officer, Platoon Leader**Fort Benning, GA*

- Second in command of a 198 Soldier Engineer Company responsible for U.S. Army Corps-level fixed, non-fixed, and float bridging; managed the company's operations, training, administrative, and maintenance processes.
- In charge of a 64 Soldier Engineer Platoon capable of conducting all U.S. Army fixed, non-fixed, and float bridging; responsible for accountability and maintenance of equipment valued in excess of \$20 million.

PRESENTATIONS AND PUBLICATIONS

Kirschenman, W., Heese, H.S., King, R., McConnell, B. (2024). "The 2-D Bin Packing Problem with Multiple Levels of Prioritization: A Spatial Optimization Perspective." Poster presentation, Graduate Student Research Symposium, NC State University, Raleigh, NC. (Link)

Kirschenman, W., Heese, H.S., King, R., McConnell, B. (2024). "Automated Vessel Selection and Combat Load Planning." Presentation at the Annual Meeting of the Institute for Operations Research and the Management Sciences (INFORMS), Seattle, WA. (Link)

Kirschenman, W., McConnell, B., King, R. (2024). Automated Vessel Selection and Combat Load Planning. *Army Sustainment*, Vol 56, No 4, 58-61. (Link)

Kirschenman, W., Heese, H.S., King, R., McConnell, B. (2024). "Automated Vessel Selection and Combat Load Planning." Presentation at the 2024 Army Operations Research Symposium (AORS) Symposium, Aberdeen Proving Ground, MD. (Link)

Kirschenman, W., Brown, K. (2022). "Distributed Wargaming Visualization and Data Collection." Presentation at the 2022 Military Operations Research Society (MORS) Symposium, Marine Corps Base Quantico, VA. (Link)

Azari, D., **Kirschenman, W.** (2022). "Introducing Effective Information Environment Organization Strategies (EIEIOS) through GitLab for Non-Programmers." Accepted Abstract at the 2022 Military Operations Research Society (MORS) Symposium, Marine Corps Base Quantico, VA. (Link)

Kirschenman, W. (2022). Presented a paper at the 2021 Army Operations Research Symposium Best Paper Competition for wargaming on "Distributed Wargaming Visualization and Data Collection." (Link)

Kirschenman, W., Brown, K. (2021). "Distributed Wargaming Visualization and Data Collection." Presentation at the 2021 Army Operations Research Symposium (Virtual). (Link)

SELECTED MILITARY AWARDS

Bronze Star Medal (1 Oak Leaf Cluster)	2013, 2016
Meritorious Service Medal (1 Oak Leaf Cluster)	2018, 2023
Army Commendation Medal (1 Oak Leaf Cluster)	2014, 2015
Army Achievement Medal (4 Oak Leaf Clusters)	2016, 2018, 2020, 2021, 2022
Military Outstanding Volunteer Service Medal	2012
Steel de Fleury Medal (Army Engineer Association)	2018

COMMUNITY SERVICE

Cub Scout Leader, Pack 79 (White Sands Missile Range, NM)	2021 - 2022
Guest Pianist, Joyful Baptist Church (Colorado Springs, CO)	2016 - 2018
Main Pianist, Cascade Hills Church (Columbus, GA)	2011 - 2013